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The Actor Model as a Load Testing Framework

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An analogy

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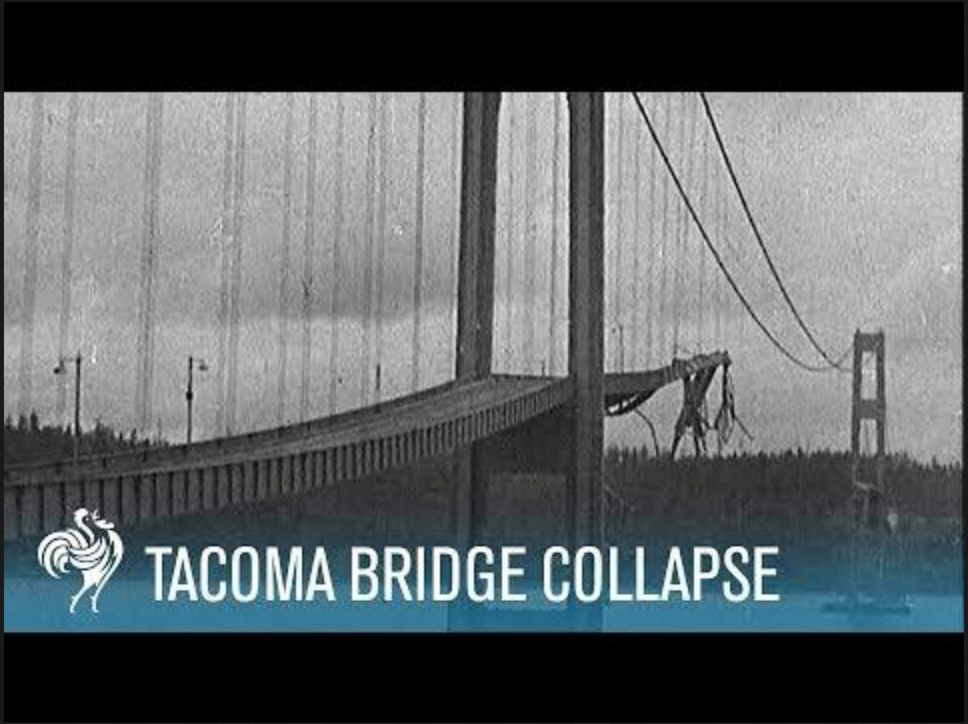
The Tacoma Bridge



The Tacoma Bridge

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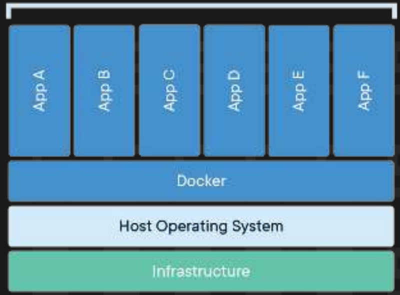
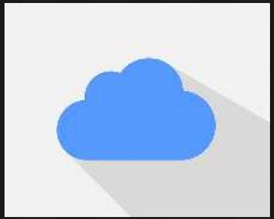
The (virtual) Tacoma Bridges





The (virtual) Tacoma Bridges

- Interactions?
- Traffic capacity?
- Amplifying factor?
- Old forces that only now start to matter?

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Terminology

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Framework

- A framework is a generic term commonly referring to an essential supporting structure which other things are built on top of. [Wikipedia]
- A supporting structure around which something can be built [Cambridge dictionary]
- A system of rules, ideas, or beliefs that is used to plan or decide something [Cambridge dictionary]

Framework

A system of rules, ideas, or beliefs that is used to plan or decide something

[Cambridge dictionary]

Model

A system of postulates, data, and inferences presented as a mathematical description of an entity

[Merriam-Webster]

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Testing

The process of using or trying something to see if it works, is suitable, obeys the rules, etc. [Cambridge dictionary]

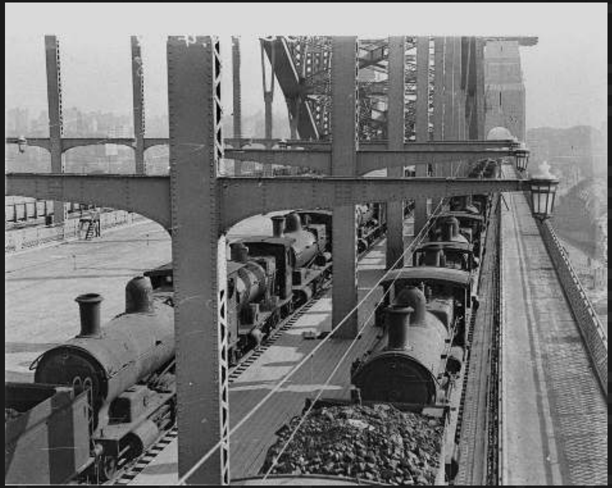
Load

A mass or quantity of something taken up and carried, conveyed, or transported [Thesaurus]

Load Testing

The process of trying something can carry a mass or quantity of work, and verify how it behaves under varying such quantities

- Performance — at minimum load
- Load — at very high load
- Stress — at failure

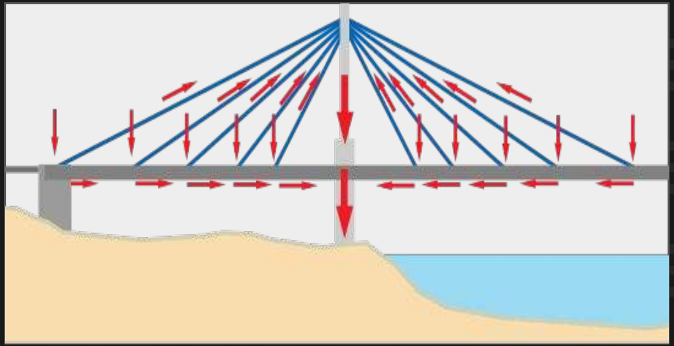


Load Testing Framework

A system of tools and ideas to apply loads in all the possible ways the system *allows*:

- Unit of measurement
- Interactions

The loads



The forces

The users

Actors

An Actor can, in response to a message it receives:

- send a finite number of messages to other Actors
- create a finite number of new Actors
- designate the behavior to be used for the next message it receives

The universal primitive

Everything is a...

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Light thickens, and the crow/ Makes wing to the rooky wood.

Shakespeare, Macbeth, Act 3, Scene 2

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A Murder of Crows

- Interactions?
- Traffic capacity?
- Amplifying factor?
- Old new forces?

Run all the actors

- *init* a scenario, once
    - Metrics, conditions, databases
- *Start* all the actors, every single time
    - Each actor executes its force
- Run the scenario
    - Locally or distributed

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Throttle

Set a Rate per Interval for an action

- Progressive rate control
- Actors wait for throttler’s approval
- Actors ask other actors to wait for throttler’s approval

Coordinate

- Pick a number of users for an action
  - all
  - a list
  - a set of distinct pairs
- Users are given to a callback as they join the coordination plan


EXAMPLE
TIME
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Match’em!

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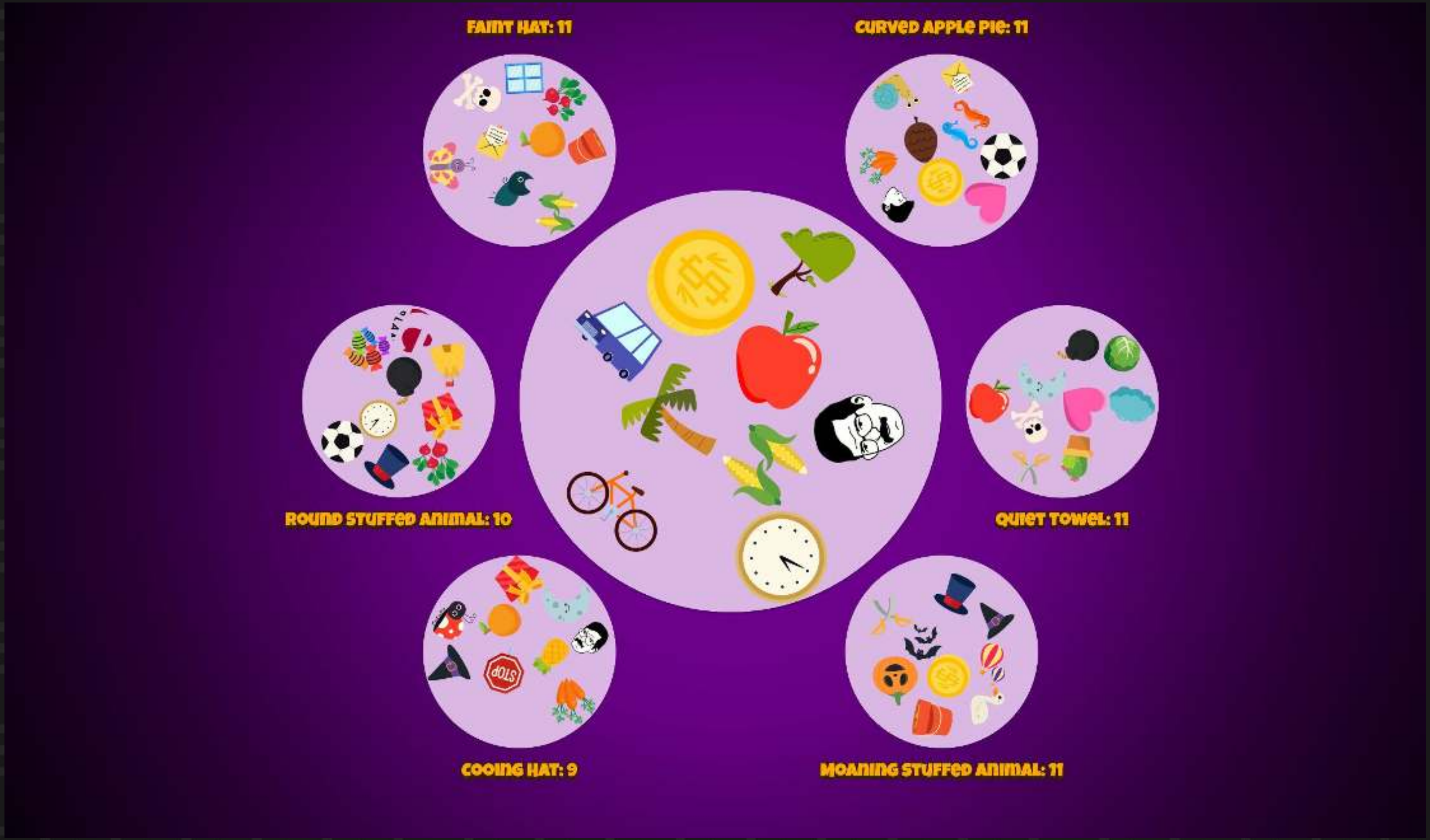
Match'em!


FAINT HAT: 11
CURVED APPLE PIE: 11
ROUND STUFFED ANIMAL: 10
QUIET TOWEL: 11
COOING HAT: 9
MOANING STUFFED ANIMAL: 11

```erlang
%% amoc_scenario
-spec init() -> ok.
init() ->
    CoordinatorPlan = coordinator_plan(),
    amoc_coordinator:start(join_users_in_room, CoordinatorPlan),
    amoc_throttle:start(connect_user, 60),
    amoc_throttle:change_rate_gradually(connect_user, 6000, 60000, ?MINUTES(1), ?SECONDS(5), 90),
    amoc_throttle:start(create_and_start_game, 60),
    amoc_throttle:change_rate_gradually(create_and_start_game, 600, 60000, ?MINUTES(1), ?SECONDS(5), 90),
    start_metrics(),
    ok.
```

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```erlang
coordinator_plan() ->
    DeckSize = amoc_config:get(set_deck_size),
    Rounds = amoc_config:get(rounds_per_game),
    UsersPerGame = amoc_config:get(users_per_game),
    [
     {UsersPerGame, fun(_Event, Users) ->
                        [FirstPid | OtherPids] = [ Pid || {Pid, _} <- Users],
                        Msg = {create_and_start_game, <<"freeForAll">>, OtherPids, DeckSize, Rounds},
                        amoc_throttle:send(create_and_start_game, FirstPid, Msg)
                    end}
    ].
```

```erlang
-spec start(amoc_scenario:user_id()) -> any().
start(MyId) ->
    amoc_throttle:send_and_wait(connect_user, waiting_until_allowed_to_connect),
    Host = amoc_config:get(host),
    UserName = integer_to_binary(MyId),
    {Delay, Jitter} = get_difficulty(),
    User = #user{host = Host, user_name = UserName,
                 delay = Delay, jitter = Jitter},
    amoc_coordinator:add(join_users_in_room, UserName),
    gen_statem:enter_loop(mm_gamer, [], initial_state, User, [{next_event, internal, init_ws}]).
```

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```erlang
coordinator_plan() ->
    DeckSize = amoc_config:get(set_deck_size),
    Rounds = amoc_config:get(rounds_per_game),
    UsersPerGame = amoc_config:get(users_per_game),
    [
     {UsersPerGame + 1, fun(_Event, Users) ->
                                UserPids = [ Pid || {Pid, _} <- Users],
                                {Group1, Group2} = lists:split(rand:uniform(length(Users)), UserPids),
                                create_and_start_game(Group1, DeckSize, Rounds),
                                create_and_start_game(Group2, DeckSize, Rounds)
                        end}
    ].

create_and_start_game([FirstPid | OtherPids], DeckSize, Rounds) ->
    Msg = {create_and_start_game, <<"freeForAll">>, OtherPids, DeckSize, Rounds},
    amoc_throttle:send(create_and_start_game, FirstPid, Msg).
```

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README.md

A Murder of Crows

CI passing

A Murder of Crows, aka amoc, is a simple framework for running massively parallel tests in a distributed environment.

It can be used as a rebar3 dependency:

```
{deps, [
    {amoc, {git, "https://github.com/esl/amoc", {tag, "2.0.1"}}}
]}.
```

MongooseIM is continuously being load tested with Amoc. All the XMPP scenarios can be found here.

In order to implement and run locally your scenarios, follow the chapters about developing and running a scenario locally. Before setting up the distributed environment, please read through the configuration overview. If you wish to run load tests via http api, take a look at the REST API chapter.

https://github.com/esl/amoc

https://github.com/esl/amoc/pull/145

README.md

amoc_arsenal_xmpp

This is a collection of XMPP-related scenarios for amoc, that Erlang Solutions uses to load test MongooseIM - our robust, scalable and efficient XMPP server.

https://github.com/esl/amoc-arsenal-xmpp

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esl/MongooseIM

esl/amoc

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